

Kafka Broker Disk Full SOP

Objective

Recover Kafka brokers that stop or degrade because broker disk utilization reaches critical levels.

Scope

Applicable to Kafka-based applications, streaming platforms, backend services, data engineering pipelines, DevOps teams, and platform teams supporting production or synthetic incident workflows.

Pre-requisites

- 1. Access to Kafka cluster, namespace or broker host.
- 2. Access to monitoring dashboards such as Grafana, Prometheus, or cloud monitoring.
- 3. Kafka CLI tools: kafka-topics, kafka-consumer-groups, kafka-console-producer, and kafka-console-consumer.
- 4. Permission to view application logs, broker logs, and deployment status.
- 5. Escalation contact for DevOps, platform, and application owner teams.

Incident Metadata

Field	Value
Ticket ID	T038
Issue	Kafka broker disk full
Category	Infrastructure
Service	Kafka
Severity	Critical
Status	Resolved
Description	Broker stopped due to disk full
Expected Resolution	Clean logs or increase disk capacity

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
L015	5/1/2026 10:07	CRITICAL	Disk full on broker	KAFKA_DISK
L035	5/1/2026 10:17	CRITICAL	Disk full on broker	KAFKA_DISK

Procedure

- 1. Confirm disk full alert and identify affected broker and mount path.
- 2. Check broker status, disk usage, and largest Kafka log directories.
- 3. Review retention settings and topic log size by partition.
- 4. Do not manually delete active Kafka segment files without approval.
- 5. Increase disk capacity or clean old data using approved retention configuration.
- 6. Restart broker if it stopped due to disk exhaustion.
- 7. Verify partitions return online and replication catches up.
- 8. Implement preventive alerts and update incident ticket with disk metrics.

Common Commands

```
kafka-topics.sh --bootstrap-server <broker>:9092 --describe
kafka-consumer-groups.sh --bootstrap-server <broker>:9092 --describe --group <group>
```

```
kafka-broker-api-versions.sh --bootstrap-server <broker>:9092
kubectl get pods -n <namespace> | grep kafka
kubectl logs <kafka-pod> -n <namespace> --tail=100
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Disk full	Retention too high or traffic spike	Increase disk or reduce retention
Broker stopped	No space left on device	Free approved space and restart broker
Replication slow	Broker recovering large logs	Monitor under-replicated partitions
Disk fills again	Uncontrolled topic growth	Review topic retention and producers

Best Practices

1. Alert before disk reaches critical threshold.
2. Use topic-level retention policies.
3. Capacity plan by ingress rate and retention.
4. Avoid manual file deletion unless guided by Kafka admin process.

References

Apache Kafka Docs | Internal Logs CSV | Tickets CSV | Monitoring Dashboards | Kafka CLI Runbook